

Math 103

Class 12

1. Two events are mutually exclusive if they cannot occur at the same time. (They have no outcomes in common.)
2. When two events are mutually exclusive, the probability that A or B will occur is $P(A \text{ or } B) = P(A) + P(B)$. (Addition Rule #1)

If A and B are not mutually exclusive, then

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

3. Example:

Determine whether the following events are mutually exclusive.

- a.) Roll a die: Get an even number, and get a number less than 3. (not)
 - b.) Roll a die: Get a prime number (2,3,5) and get an odd #. (not)
 - c.) Roll a die: Get a number greater than 3 and get a number less than 3. (yes)
 - d.) Select a student in your class: The student has blond hair and the student has blue eyes. (not)
 - e.) Select a student in your college: The student is a sophomore and the student is a business major. (not)
 - f.) Select any course: It is a calculus course and it is an English course. (yes)
 - g.) Select a registered voter: The voter is a Republican and the voter is a Democrat. (yes)
4. Example: In a large department store, there are 2 managers, 4 department heads, 16 clerks, and 4 stock persons. If a person is selected at random, find the probability that the person is either a clerk or a manager.

26 people

$$\text{Clerk or Manager: } 16/26 + 2/26 = 18/26 = 9/13$$

5. Example: An automobile dealer has 10 Fords, 7 Buicks, and 5 Plymouths on her used-car lot. If a person purchases a used car, find the probability that it is a Ford or Buick.

22 cars total

$$10/22 + 7/22 = 17/22$$

6. The probability of a California teenager owning a surfboard is 0.43, of owning a skateboard is 0.38, and of owning both is 0.28. If a California teenager is selected at random, find the probability that he or she owns a surfboard or a skateboard.

$$P(A) + P(B) - P(A \text{ and } B) \\ 0.43 + 0.38 - 0.28 = 0.53$$

7. A single card is drawn from a deck. Find the probability of selecting the following:

- a.) a 4 or a diamond
- b.) a club or a diamond
- c.) a jack or a black card

$$a.) \quad P(4) + P(D) - P(4 \text{ and } D) \\ 4/52 + 13/52 - 1/52 = 16/52 = 4/13$$

$$b.) \quad P(C) + P(D) = \\ 1/4 + 1/4 = 1/2$$

$$c.) \quad P(J) + P(\text{Black}) - P(\text{Black and Jack}) \\ 4/52 + 26/52 - 2/52 = 28/52 = 7/13$$

8. A large department store purchases 400 boomboxes for a sale. Two hundred fifty of them have CD players and 150 of them have cassette players: 140 of the boomboxes with CD players are black, and the rest are white; 80 of the boomboxes with cassette players are black, and the rest are white. If a boombox is selected at random, find the probability that it will

- a.) be black or have a CD player.
- b.) be white or have a cassette player.

	CD Players	Cassette	Totals
Black	140	80	220
White	110	70	180
Totals	250	150	400

$$a.) \quad \text{Black or CD} - (\text{Black \& CD}) \\ 220/400 + 250/400 - 140/400 = 330/400 = 33/40$$

$$b.) \quad \text{White or Cassette} - (\text{White \& Cassette}) \\ 180/400 + 150/400 - 70/400 = 260/400 = 26/40 = 13/20$$

9. In a certain geographic region, newspapers are classified as being published daily morning, daily evening, and weekly. Some have a comics section and others do not. The distribution is shown here.

Have comics section	Morning	Evening	Weekly	Total
Yes	2	3	1	6
No	3	4	2	9
Total	5	7	3	15

If a newspaper is selected at random, find these probabilities:

- a.) The newspaper is a weekly publication.

$$3/15 = 1/5$$

- b.) The newspaper is a daily morning publication or has comics.

$$P(\text{morning}) + P(\text{comics}) - P(\text{morning with comics})$$

$$5/15 + 6/15 - 2/15 = 9/15 = 3/5$$

- c.) The newspaper is published weekly or does not have comics.

$$P(\text{weekly}) + P(\text{no comics}) - P(\text{weekly and no comics})$$

$$3/15 + 9/15 - 2/15 = 10/15 = 2/3$$

10. Ashley is playing a board game. On her next move she needs to throw a sum larger than 8 on the two dice in order to win. What is the probability that she will roll a sum bigger than 8?

She must roll a sum of 9, 10, 11 or 12.

The probability of rolling a sum greater than 8 is the same as P(9 or 10 or 11 or 12)

Since the events are mutually exclusive,

$$\begin{aligned}
 P(9 \text{ or } 10 \text{ or } 11 \text{ or } 12) &= P(9) + P(10) + P(11) + P(12) \\
 &= 4/36 + 3/36 + 2/36 + 1/36 \\
 &= 10/36 \\
 &= 5/18
 \end{aligned}$$

11. At Findlay Electronics, all 140 employees were asked about their political affiliations. The employees were grouped by type of work, as executives or production workers.

Employee Type	Political Affiliation			Total
	Democrat (D)	Republican (R)	Independent (I)	
Executive (E)	5	34	9	48
Production worker	63	21	8	92
Total	68	55	17	140

- a.) Find $P(D)$ $68/140 = .486$
 b.) Find $P(E)$ $48/140 = .343$
 c.) Find $P(D|E)$ $5/48 = .104$
 d.) Find $P(D \text{ and } E)$ $5/140 = .036$
 e.) Find $P(D \text{ or } E)$ $111/140 = .793$
 $68/140 + 48/140 - 5/140 = 111/140$